A PROJECT REPORT
ON

Web-Based Editor

Submitted in partial fulfillment of the requirement for the award of Degree of Bachelor of Technology in Computer Science & Engineering

Submitted to:

Rajasthan Technical University, Kota (Raj.)

Submitted By:

Satya Raj Prakash (21EARCS147)
Vivek Jha (21EARCS182)
Shreya Kejriwal (21EARCS155)
Yash Raushan (21EARCS186)

Under the supervision of

GUIDE
Dr. Vibhakar Pathak
Professor

PROJECT COORDINATOR
Dr. Vishal Shrivastava
Professor

Session: 2024-25

Department of Computer Science & Engineering
ARYA COLLEGE OF ENGINEERING & I.T.
SP-42, RIICO INDUSTRIAL AREA, KUKAS, JAIPUR

Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the work embodies in this Project entitled "**Web-Based Editor**" being submitted by "**Satya Raj Prakash (21EARCS147), Vivek Jha (21EARCS182), Shreya Kejriwal (21EARCS155) and Yash Raushan (21EARCS186)"** in partial fulfillment of the requirement for the award of "**Bachelor of Technology in Computer Science & Engineering"** to Rajasthan Technical University, Kota (Raj.) during the academic year 2024-25 is a record of bonafide piece of work, carried out by them under our supervision and guidance in the **"Department of Computer Science & Engineering", Arya College of Engineering & Information Technology, Jaipur**.

Project Guide
Dr. Vibhakar Pathak
Professor

Project-coordinator
Dr. Vishal Shrivastava
Professor

Approved by

Dr. Akhil Pandey
Head, Department of
Computer Science & Engineering

Department of Computer Science & Engineering

CERTIFICATE OF APPROVAL

The Project Report entitled "**Web-Based Editor**" submitted by "Satya Raj Prakash (21EARCS147), Vivek Jha (21EARCS182), Shreya Kejriwal (21EARCS155), Yash Raushan (21EARCS186)" has been examined by us and is hereby approved for carrying out the project leading to the award of degree "**Bachelor of Technology in Computer Science & Engineering"**. By this approval the undersigned does not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn therein, but approve the pursuance of project only for the above mentioned purpose.

Project Guide
Dr. Vibhakar Pathak
Professor

Project Coordinator
Dr. Vishal Shrivastava
Professor

Department of Computer Science & Engineering

DECLARATION

We "**Satya Raj Prakash, Vivek Jha, Shreya Kejriwal and Yash Raushan**", students of **Bachelor of Technology** in **Computer Science & Engineering, session 2024-25, Arya College of Engineering & Information Technology, Jaipur**, here by declare that the work presented in this Project entitled "**Web-Based Editor**" is the outcome of our own work, is bonafide and correct to the best of our knowledge and this work has been carried out taking care of Engineering Ethics. The work presented does not infringe any patented work and has not been submitted to any other University or anywhere else for the award of any degree or any professional diploma.

Satya Raj Prakash (21E1ARCSM45P147)
Vivek Jha (21E1ARCSM45P182)
Shreya Kejriwal (21E1ARCSF40P155)
Yash Raushan (21E1ARCSM45P186)

Date: 13-05-2025

Department of Computer Science & Engineering

ACKNOWLEDGEMENT

We take this opportunity to express our heartfelt gratitude to all those who have supported us throughout the development of our project titled ***Web-Based Editor*** – *Interface Used to integrate services for Web based application*.

First and foremost, we extend our sincere thanks to our respected project guide, **Dr. Vibhakar Pathak**, for their continuous encouragement, expert guidance, and valuable suggestions throughout the duration of the project. Their consistent support and mentorship were vital in shaping our ideas and turning them into a successful working prototype.

We are also extremely thankful to the **Head of the Department, Dr. Akhil Pandey**, for providing us with the required resources, motivation, and a positive academic environment that greatly contributed to the smooth execution of this project.

Our sincere thanks also go to all the **faculty members of the Computer Science Department** for their insightful feedback and help at various stages of our project development. Their academic input helped us in better understanding the theoretical and practical aspects of the project.

We are grateful to our **college management** for offering a conducive infrastructure, research environment, and facilities necessary for the completion of this project.

Last but not least, we appreciate the cooperation and teamwork of our fellow project members. The dedication and effort of every individual involved played a significant role in the successful completion of ***Web-Based Editor***.

Department of Computer Science & Engineering

ABSTRACT

The **Web-Based Editor** project presents an intelligent and interactive platform that bridges the gap between algorithmic thinking and actual programming through a browser-accessible, AI-integrated code editing environment. While traditional code editors require users to be well-versed in programming syntax and development tools, this system empowers a broader range of users—including students, educators, non-programmers, and developers—by enabling them to convert logic or plain language into structured, executable code automatically.

The platform is developed using **React.js** for the frontend and **Node.js** with **Express.js** for the backend, ensuring a responsive and modular structure suitable for web deployment. The highlight of this editor is its integration with **Hugging Face's AI models**, which utilize natural language processing (NLP) to interpret pseudocode or plain English instructions and translate them into actual programming code. This approach not only reduces the complexity of writing syntactically correct code but also supports users who are still learning the fundamentals of programming.

To further enhance user interaction and learning, the Web-Based Editor includes a **real-time live preview section**, which renders the generated code output instantly in the browser. This feature allows immediate visual feedback, making it easier to understand how logical inputs are transformed into functioning code. Such feedback is crucial for educational purposes, debugging, and iterative development.

The system is particularly beneficial for academic environments, where it can help instructors demonstrate concepts across multiple programming languages, and for rapid prototyping by developers who need to quickly visualize and test code snippets. It also provides a gateway for non-technical users to describe functionality in natural language and receive working code without having to master syntax.

This project aims to democratize programming by reducing entry barriers, promoting self-paced learning, and accelerating early-stage software development. With its modular design, scalable architecture, and extensible AI capabilities, the Web-Based Editor lays the groundwork for future low-code or no-code development tools. It is a step forward in making intelligent, accessible, and efficient coding platforms a reality for users of varying skill levels

TABLE OF CONTENT

S. No.	Content	Page No.
1.	Objective & Scope of the Project	1 – 3
2.	Funding Details (With Proof)	4 – 7
3.	Theoretical Background	8 – 13
4.	Definition of the Problem	14 – 16
5.	System Analysis & User Requirements	17 – 24
6.	System Planning (PERT Chart)	25 – 30
7.	Methodology Adopted & Details of Hardware and Software Used	31 – 35
8.	Detailed Life Cycle of the Project	36 – 40
9.	ERD (Entity Relationship Diagram) & DFD (Data Flow Diagram)	41 – 45
10.	Process Involved, Algorithm and Flowchart	46 – 51
11.	Input and Output Screen Design (Screenshots)	52 – 58
12.	Printout of the Code Sheet (Coding)	59 – 62
13.	Testing	63 – 67
14.	User/Operational Manual (including security aspects, access rights, backup, controls, etc.)	68 – 73
15.	Conclusions	74 – 77
16.	Future Enhancement	78 – 82
17.	References	83